

SARTHAK DRAVID

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 LinkedIn |  GitHub

Education

Manipal Institute of Technology (MAHE)

Jul. 2023 – Jul. 2027

B.Tech in Cyber Physical Systems

CGPA: 7.9/10

Relevant Coursework: Data Structures and Algorithms; Computer Architecture and Organization; Data Communication Networks; Unsupervised Intelligence in CPS; Sensor Technology; Augmented and Virtual Reality

Experience

Nakshatra Mapping Solutions

Aug. 2024 – Dec. 2024

Innovation Team Member

- Led research across AR, 3D mapping, and immersive systems, including visual SLAM implementation and advanced 3D reconstruction work.
- Represented the innovation roadmap, research progress, and IP strategy in investor and partner discussions.
- Worked directly with clients to design immersive AR experiences for real-world use cases.

Open Horizon

Feb. 2025 – Present

3D Engineer

- Worked on 3D reconstruction research focused on mapping educational institutes at high fidelity.
- Created high-resolution building models using Meshroom and Blender and contributed to the open-source [MapMIT](#) initiative.
- Assisted in the development of a fully open-source autonomous drone.

Freelance

May 2025 – Jul. 2025

CAD Designer

- Developed professional-grade technical CAD drawings tailored to client specifications.
- Applied 3D reconstruction workflows to model client prototypes with greater precision.

Projects

Guided Tours

- Built a comprehensive AR museum-tour solution combining indoor navigation with context-aware descriptions.
- Designed a scalable node-based navigation system using Three.js and WebAR, with an admin platform to upload indoor maps, assign nodes, and export graph data as GeoJSON.
- Built a WebAR client to ingest exported node graphs and render live navigation paths, waypoints, and image-tracked 3D information panels with stable multi-marker performance.
- Optimized mobile-browser rendering with Troika SDF text, Vite 6, and HTTPS-based local hosting for low-latency spatial interaction.

Universal Deep Reinforcement Learning Agent for Multi-Scenario CSTR Temperature Control

- Developed a universal temperature-control agent using TD3 for a jacketed continuous stirred-tank reactor.
- Implemented a 5-bit scenario mask in the state vector so a single policy network could learn seven actuator configurations.
- Analyzed manipulated variables to identify feed-stream temperature as the dominant control authority and built a Streamlit dashboard for real-time performance visualization.

PC Snapshot

- Created a Windows desktop application with React, TypeScript, and Vite to capture and restore full workspace states.
- Built an automated capture engine that scans active processes and stores contextual metadata from IDEs, browsers, and terminals.
- Implemented a sequential restoration workflow that relaunches applications, replays context macros, and repositions windows to saved coordinates.

Cricket DRS Outcome Prediction System

- Built an end-to-end predictive system to model DRS outcomes by converting natural-language match commentary into structured physical trajectory features.
- Designed an NLP-to-physics pipeline using spaCy and BERT/RoBERTa-style transformers to reconstruct ball-tracking signals such as pitch coordinates and impact zones.
- Combined XGBoost and LightGBM classifiers with Bayesian uncertainty quantification and Win Probability Added to estimate both review success probability and challenge value in real time.
- Deployed the system with a FastAPI backend and Streamlit dashboard for tactical decision support and heatmap-based visualization.

Technical Skills

Languages	Python, R, C++, C#, JavaScript (Three.js), SQL (Basic)
Data Science & AI	NumPy, Pandas, TensorFlow, PyTorch, OpenCV, NLP, Snowflake
XR & 3D Development	WebXR, Three.js, Unity, Blender, vSLAM, Meshroom
Tools	Git/GitHub, GeoJSON, MATLAB/Simulink, QGIS, AutoCAD, Vite
Graphics	Photoshop, Krita, Canva, L ^A T _E X, Mermaid
Hardware	Raspberry Pi, Arduino, 8051, ARM
Networking	Cisco Packet Tracer, Wireshark

Certifications

- [Python for Data Analysis: Pandas & NumPy](#)
- [Wireless Communications Onramp by MATLAB](#)
- [Signal Processing Onramp by MATLAB](#)
- [Machine Learning for Data Science Projects by IBM](#)
- [Exploratory Data Analysis by Johns Hopkins University](#)
- [R Programming by Johns Hopkins University](#)

Achievements & Extracurricular

- [Ground Reality, BITS Hyderabad 2025](#): Won Best Startup Pitch with a cash prize of INR 30,000 after competing with 250+ teams across three rounds over two months.
- Successfully launched [MapMIT](#), receiving 20k requests within the first two weeks of launch.
- Received the [Emergent Ventures Fellowship Grant](#) by the Mercatus Center at George Mason University.
- Reached the top 3% of the [WTF Fund C1/25 cohort](#).
- Mars Rover Manipal: Junior task-phase member, Research AI; AAINA Dramatics: Sets and Design Head.